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RadeonSI

RadeonSI is the official, open-source OpenGL graphics driver for modern AMD Radeon graphics cards on Linux operating systems. It is developed by AMD engineers alongside the open-source community and is distributed as a core part of the [Mesa 3D Graphics Library](#).

Architecture & Support

Unlike older legacy drivers, RadeonSI is built on the **Gallium3D** architecture. It acts as the user-space driver handling OpenGL instructions, while relying directly on the **AMDGPU** Linux kernel driver to communicate with the hardware. Phoronix +2

- **Hardware Compatibility:** It natively supports graphics cards based on the **Graphics Core Next (GCN)** architecture (such as the Radeon HD 7000 series) all the way up to the latest **RDNA** architectures (including RDNA4/GFX12) and integrated APUs. Phoronix +2
- **Shader Compiler:** Historically, RadeonSI relied on the LLVM compiler infrastructure. However, the driver transitioned to using **ACO** (the Valve-backed AMD Compiler) by default. This change drastically minimizes OpenGL Shading Language (GLSL) compilation times, reduces micro-stuttering, and speeds up video game loading times. Phoronix +2

Key Technical Milestones

RadeonSI has continually evolved to match modern gaming and application standards:

- **Unified Video Decoding:** The multimedia decoding architecture is unified between RadeonSI and the RADV (Vulkan) driver, streamlining GPU-

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Other AMD Technologies for Mesa or Gallium3D ?

Beyond RadeonSI, AMD's open-source presence in **Mesa** and the **Gallium3D** architecture includes several critical drivers and specialized subsystems. Together, these technologies handle Vulkan gaming, video encoding/decoding, compute workloads, and legacy hardware support on Linux.

1. RADV / The Open Source Vulkan Driver



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standard for AMD Linux gaming. AMD natively supports it alongside Valve engineers.  Phoronix +2

- **Ray Tracing & Proton Compatibility:** RADV handles translation layers like DXVK and VKD3D (which map DirectX 9/11/12 to Vulkan in Steam Play/Proton), and it features mature Ray Tracing pipelines.  YouTube · Brodie Robertson

2. ACO (The AMD Compiler)

Originally introduced by Valve specifically for RADV, **ACO** is a shader compiler backend that bypasses the massive LLVM infrastructure.

- ACO optimizes shader compilation speeds, directly resulting in faster game load times and eliminating the infamous "shader compilation stutter" during gameplay.
- After proving highly successful in RADV, ACO integration has steadily expanded across other pieces of the Mesa ecosystem.

3. Rusticl (Modern Compute / OpenCL)

In the Gallium3D architecture, compute workloads are handled via specialized "State Trackers." Historically, AMD relied on *Clover* for OpenCL, but Mesa has completely transitioned to **Rusticl**.  timur.hu

- Rusticl is a modern, open-source OpenCL implementation written in Rust.
- It works as a Gallium state tracker, translating OpenCL API calls into an Intermediate Representation (NIR) that RadeonSI or other Gallium drivers can compile for the GPU hardware.  Wikipedia +1

4. Multimedia State Trackers (VA-API & VDPAU)

Gallium3D handles hardware-accelerated video decoding and encoding using dedicated state trackers that interface directly with the hardware:





control AMD's **UVD/VCE** (Unified Video Decoder / Video Coding Engine on older cards) or **VCN** (Video Core Next on modern RDNA cards) silicon blocks.  Phoronix

5. Legacy & Vintage Gallium3D Drivers

Before RadeonSI was created for GCN architecture and newer, older generations of ATI/AMD hardware were split into different Gallium3D drivers. These are still actively maintained by community enthusiasts to keep vintage hardware functional:

- **R600g**: A Gallium3D driver supporting the Radeon HD 2000 through HD 6000 series. It has undergone code cleanups to modernize its infrastructure.
- **R300g**: A remarkably resilient driver supporting the ancient ATI R300 through R500 (Radeon X1000) chips. Developers overhauled R300g to strip out legacy intermediate frameworks (TGSI) and adopt modern NIR compilers, keeping 20+ year-old hardware compatible with modern Linux software stacks.  Phoronix +3

Summary Layout: How They Connect

Technology	Role	Layer / Architecture
RadeonSI	OpenGL / OpenGL ES	Gallium3D Driver
RADV	Vulkan (Gaming / Proton)	Mesa Vulkan Driver
Rusticl	OpenCL (Compute Architecture)	Gallium3D State Tracker
VA-API / VVDALL	Video Encoding & Decoding	Gallium3D Multimedia Tracker





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